WORKED SOLUTION

STEP II 1989, Question 13

Mechanics • Moments of Inertia

STEP II 1989, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q13

This question has a full guided treatment in *Beyond STEP*, Section 11.7, *Simpson's Rule in a Rod*: https://beyondstep.keithmcnulty.org/ch11/simpsons-rod.html

The key idea

Everything rests on two facts about moments of inertia. The parallel axis theorem says the moment of inertia about any axis is fixed by the mass, the centre of mass and the moment of inertia about the parallel axis through the centre of mass. And moments of inertia are *additive*, so a body can be cut into pieces and each piece swapped for something equivalent. A lamina is a stack of thin rods, so once we know how to replace one rod, we can replace the lamina.

Solution

Equivalence about every axis

Let ℓ be any axis, and let ℓ_O be the parallel axis through O, a distance d from ℓ. Both the body and the particle system have mass m and centre of mass O, so we can apply the parallel axis theorem to each of them:

$$I_\ell^{\text{body}} = I_{\ell_O}^{\text{body}} + md^2, \qquad I_\ell^{\text{particles}} = I_{\ell_O}^{\text{particles}} + md^2.$$

By hypothesis $I_{\ell_O}^{\text{body}} = I_{\ell_O}^{\text{particles}}$, because ℓ_O passes through O. So the two right-hand sides are equal, and $I_\ell^{\text{body}} = I_\ell^{\text{particles}}$ for every axis ℓ, as required.

The uniform rod

Put the rod along the x-axis from $x = -a$ to $x = a$, so its centre of mass O is the origin. Place particles of masses p, q, r at $x = -a$, 0, a. For total mass m and centre of mass at

O we need

$$p + q + r = m, \qquad -pa + ra = 0,$$

so $r = p$ and $q = m - 2p$.

Now take any axis through O making an angle θ with the rod. A point of the rod at position x is a distance $|x| \sin\theta$ from this axis. So every moment of inertia about an axis through O is $\sin^2\theta$ times $\sum(\text{mass})\, x^2$, and we only need that one sum to agree. For the rod, with mass $\frac{m}{2a}\,\mathrm{d}x$ in each element,

$$\int_{-a}^{a} \frac{m}{2a}\, x^2 \,\mathrm{d}x = \frac{m}{2a} \cdot \frac{2a^3}{3} = \tfrac{1}{3}ma^2,$$

and for the particles it is $pa^2 + q \cdot 0 + pa^2 = 2pa^2$. These agree for every θ when $p = \frac{1}{6}m$. So the rod is dynamically equivalent to

$$\tfrac{1}{6}m \text{ at each end}, \qquad \tfrac{2}{3}m \text{ at the midpoint}.$$

By the first part, the two then have the same moment of inertia about *any* axis, not only axes through O.

The rectangular lamina

Let the rectangle have centre O and sides $2a$ (parallel to the x-axis) and $2b$, so its mass per unit area is $M/(4ab)$. Cut it into thin strips parallel to the x-axis. The strip at height y with width δy is a uniform rod of length $2a$ and mass $\delta M = \frac{M}{2b}\,\delta y$.

> **Ask yourself** The centre of a strip isn't at O. Why may we still replace each strip by its three particles?

This is exactly what the first part is for. Each strip has the same moment of inertia as its three particles about *every* axis, not just axes through the strip's own centre. Moments of inertia add, so the lamina has the same moment of inertia, about any axis, as the collection of all the replacement particles. That collection also has total mass M and centre of mass O, because each strip's replacement has the same mass and centre of mass as the strip.

The strip at height y is replaced by $\frac{1}{6}\delta M = \frac{M}{12b}\,\delta y$ at each of $(\pm a, y)$ and $\frac{2}{3}\delta M = \frac{M}{3b}\,\delta y$ at $(0, y)$. As y runs from $-b$ to b, these particles fill out three uniform rods of length $2b$:

- one along each side $x = \pm a$, each of mass $\frac{M}{12b} \cdot 2b = \frac{1}{6}M$;
- one along the centre line $x = 0$, of mass $\frac{M}{3b} \cdot 2b = \frac{2}{3}M$.

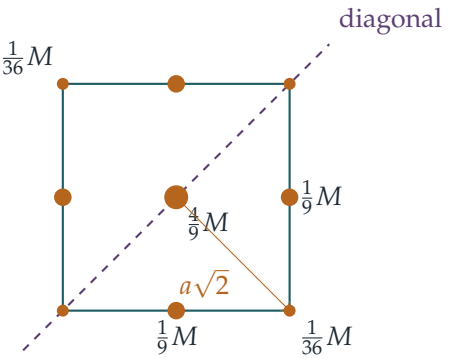
Now apply the rod result to each of these rods, by the same additivity argument:

- each side rod ($\frac{1}{6}M$) becomes $\frac{1}{6} \cdot \frac{1}{6}M = \frac{1}{36}M$ at each of its ends, which are corners, and $\frac{2}{3} \cdot \frac{1}{6}M = \frac{1}{9}M$ at its midpoint, the midpoint of a side of length $2b$;
- the central rod ($\frac{2}{3}M$) becomes $\frac{1}{6} \cdot \frac{2}{3}M = \frac{1}{9}M$ at each of its ends, the midpoints of the sides of length $2a$, and $\frac{2}{3} \cdot \frac{2}{3}M = \frac{4}{9}M$ at the centre.

So there is $\frac{1}{36}M$ at each corner, $\frac{1}{9}M$ at the midpoint of each side and $\frac{4}{9}M$ at the centre. The total is $\frac{4}{36}M + \frac{4}{9}M + \frac{4}{9}M = M$, and by symmetry the centre of mass is O. This is the required equivalent system.

The square about a diagonal

For the square of side $2a$, put the corners at $(\pm a, \pm a)$ and take the diagonal $y = x$. The distance of (x_0, y_0) from this line is $|x_0 - y_0|/\sqrt{2}$.



- The corners (a, a), $(-a, -a)$ and the centre lie on the diagonal and contribute nothing.
- The corners $(a, -a)$, $(-a, a)$ are at distance $2a/\sqrt{2} = a\sqrt{2}$.
- Each midpoint $(\pm a, 0)$, $(0, \pm a)$ is at distance $a/\sqrt{2}$.

Hence

$$I_{\text{diagonal}} = 2 \cdot \tfrac{1}{36}M \cdot 2a^2 + 4 \cdot \tfrac{1}{9}M \cdot \tfrac{1}{2}a^2 = \tfrac{1}{9}Ma^2 + \tfrac{2}{9}Ma^2 = \tfrac{1}{3}Ma^2.$$

The non-uniform rod

Let the rod have length $2a$, lying along the x-axis from $x = 0$ to $x = 2a$, with line density kx. Then

$$m = \int_0^{2a} kx \, \mathrm{d}x = 2ka^2, \qquad \text{so} \qquad k = \frac{m}{2a^2}.$$

Its centre of mass is at

$$\bar{x} = \frac{1}{m}\int_0^{2a} kx^2 \, \mathrm{d}x = \frac{k}{m} \cdot \frac{8a^3}{3} = \frac{4a}{3},$$

and

$$\int_0^{2a} kx^3 \, \mathrm{d}x = \frac{k\,(2a)^4}{4} = 4ka^4 = 2ma^2.$$

As with the uniform rod, every point is on one line, so about an axis through the centre of mass at angle θ to the rod the moment of inertia is $\sin^2\theta \sum(\text{mass})(x - \bar{x})^2$. If the masses and centres of mass agree, $\sum(\text{mass})(x - \bar{x})^2 = \sum(\text{mass})x^2 - m\bar{x}^2$, so it is enough to match $\sum(\text{mass})\,x^2$, which is the moment of inertia about the perpendicular axis through the end $x = 0$.

Let the particles be p_0 at $x = 0$, p_1 at $x = a$ and p_2 at $x = 2a$. Matching the mass, the first moment and the second moment gives the equations

$$\begin{aligned} p_0 + p_1 + p_2 &= m, \\ p_1 a + 2p_2 a &= \tfrac{4}{3}ma, \\ p_1 a^2 + 4p_2 a^2 &= 2ma^2. \end{aligned}$$

Subtracting the second (divided by a) from the third (divided by a^2) gives $2p_2 = \frac{2}{3}m$, so $p_2 = \frac{1}{3}m$. Then $p_1 = \frac{4}{3}m - \frac{2}{3}m = \frac{2}{3}m$, and finally $p_0 = m - \frac{2}{3}m - \frac{1}{3}m = 0$.

The particle at the light end has zero mass, so only two particles are needed: $\frac{2}{3}m$ at the midpoint and $\frac{1}{3}m$ at the heavy end.

Checks

- The standard result for a square lamina of side $2a$ about an axis through its centre in its plane, parallel to a side, is $\frac{1}{3}Ma^2$. By the perpendicular axis theorem and symmetry, every in-plane axis through the centre gives the same value, so the diagonal must too. This agrees.
- For the non-uniform rod, the two particles give mass m, first moment $\frac{2}{3}m \cdot a + \frac{1}{3}m \cdot 2a = \frac{4}{3}ma$ (so the centre of mass is at $\frac{4}{3}a$), and second moment $\frac{2}{3}ma^2 + \frac{4}{3}ma^2 = 2ma^2$, all as required.
- Numerically, with a fine grid of point masses, we compared each body with its particle system about 100–200 random axes in space (not through the centre). The rod and the variable rod agreed to better than 10^{-7}, the rectangle to 4×10^{-5} (grid error), and the square about its diagonal gave 0.33332 for $M = a = 1$.

Common slips

- Replacing each strip by its three particles without saying why that's allowed. The strips aren't centred at O, so you need the first part's "any axis" result.
- In the last part, imposing only the mass and centre of mass conditions. Two equations can't fix three masses: the moment of inertia condition is what forces $p_0 = 0$.

Beyond STEP

For a rigid body moving in space, what matters dynamically is its mass, its centre of mass and its *inertia tensor*, a symmetric 3×3 matrix that gives the moment of inertia about every axis through the centre of mass. Two bodies sharing these move identically under the same forces, and that is why "dynamically equivalent" is a good name. Matching the first few moments of a distribution with a few point masses is also the idea behind numerical integration. The weights $\frac{1}{6}, \frac{2}{3}, \frac{1}{6}$ for the uniform rod are exactly Simpson's rule, and choosing the positions as well as the weights leads to Gaussian

quadrature. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops the inertia tensor and rigid-body motion.

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